Xinyuan Luo

Robotics Ph.D. Student, Duke University

xl521@duke.edu | Durham, NC | Website | Google Scholar | LinkedIn

Research Interests

Humanoid whole-body control, compliance, physical interaction, and loco-manipulation; robotic system integration and agentic robotics.

Education

Duke University — Durham, NC
Ph.D. in Robotics — *2025 – Present*
Advisors: Prof. Xianyi Cheng and Prof. Boyuan Chen

University of Illinois Urbana-Champaign — Urbana, IL
M.Eng. in Autonomy and Robotics — *2023 – 2024*
Advisor: Prof. Wenzhen Yuan

Shanghai Jiao Tong University — Shanghai, China
B.Eng. in Measurement, Control Technology and Instruments — *2019 – 2023*

Selected Publications

- **X. Luo**, C. Yang, B. Chen, X. Cheng. *"CEER2: Directional and Tunable End-Effector and Root Compliance for Humanoid Loco-Manipulation." Under Review*, 2026.
- **X. Luo**, X. Chen, X. Yin, H. Wu, B. Xia, Z. Chen, J. Li, B. Chen, X. Cheng. *"CEER: Compliant End-Effector and Root Control as a Unified Interface for Hierarchical Humanoid Loco-Manipulation." IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- **X. Luo**, W. Yuan. *"A Perception-Manipulation Robotic System for Food Cutting." IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- **X. Luo***, S. Jin*, H.-J. Huang, W. Yuan. *"An Intelligent Robotic System for Perceptive Pancake Batter Stirring and Precise Pouring." IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024. Best Entertainment and Amusement Paper Finalist.

Research Experience

Duke University — Durham, NC
Graduate Research Assistant — Humanoid Whole-Body Control — *2025 – Present*

- Develop learning-based **whole-body control and loco-manipulation** systems for humanoid robots, using large-scale reinforcement learning in NVIDIA Isaac Lab and sim-to-real deployment on the Unitree G1.
- Designed a **hierarchical humanoid control architecture** that connects high-level task reasoning and skill execution with low-level whole-body control through a unified end-effector–root interface, enabling modular integration of teleoperation, programmed skills, and agentic task planning.
- Developed a unified formulation of **humanoid compliance**, including directional and tunable end-effector compliance and root-level force-response behaviors.
- Built and evaluated complete humanoid systems for contact-rich loco-manipulation tasks including writing, dragging, co-carrying, and external-force interaction.

University of Illinois Urbana-Champaign — Urbana, IL
Research Assistant — Robotic Manipulation and Food Processing — *2023 – 2024*

- Developed contact-rich robotic manipulation systems on the **UR5e** for food-processing tasks including viscous-liquid stirring, precise pouring, and cutting foods with diverse physical properties.
- Integrated **force/torque sensing** for physical interaction and online estimation of task-relevant properties, including liquid viscosity, mixture uniformity, and cutting interaction forces.
- Designed and integrated complete robotic pipelines combining perception, force-based interaction, task-level decision making, and learning-based control, resulting in first-author publications at **IROS 2024** and **IROS 2026**.

Industry Experience

Kepler Robotics Shanghai, China
Robot Learning Engineer (Full-time) *Jan. 2025 – Jul. 2025*

- Developed and deployed **whole-body teleoperation** for full-size humanoid robots, and conducted real-robot teleoperation demonstrations for **General Motors Shanghai** and **BMW Shenyang / Nanjing**.
- Developed a humanoid **force-aware dual-arm manipulation system** for industrial material loading and unloading in a **BMW manufacturing environment**, enabling reliable physical interaction with workpieces during real-world operation.
- Developed **dexterous-hand control** for manipulation tasks and improved grasp robustness through mechanical redesign and modification of dexterous-hand fingers for task-specific object geometries.

Technical Skills

Robot Learning & Control: Reinforcement Learning, PPO, Whole-Body Control, Compliance Control, Motion Tracking, Teacher–Student Training, Sim-to-Real

Software: Python, C++, PyTorch, NVIDIA Isaac Lab, MuJoCo, Crocoddyl, ROS / ROS2, Linux, Git

Hardware: Unitree G1, Full-Size Humanoid Robots, UR5e, Force/Torque Sensors, GelSight Mini, Xsens Motion Capture, PICO 4 VR, Dexterous Hands